MICHIGAN MEDICINE
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# Weight-Based Dosing Recommendations for Intravenous Antimicrobials in Obese Adult Patients

The FDA has not recognized obese patients as a special population and thus pharmacokinetic studies to determine appropriate dosing is not required in this population. A literature review was conducted to help provide guidance when calculating dosing for weight-based dosing of intravenous antimicrobials in obese patients.
**Note**: Given the paucity of data, the below recommendations cannot be considered absolute, and cases should be evaluated individually in consultation with Infectious Disease/Antimicrobial Stewardship, incorporating the severity of infection and risk of toxicity.

I. Adjusted Body Weight should be utilized when calculating dosing of the following antimicrobials in obese (BMI ≥ 30) patients. In non-obese patients, Total Body Weight should be utilized to calculate dosing.
   A. Voriconazole
   B. Daptomycin
   C. IV Ganciclovir
   D. IV Acyclovir
   E. Liposomal Amphotericin B
   F. IV Trimethoprim-sulfamethoxazole
   G. Foscarnet
   H. Colistin (maximum 300 mg total daily dose)
   I. Polymyxin B (maximum 200 mg total daily dose)
   J. Ribavirin
   K. Cidofovir
   L. Flucytosine
   M. Aminoglycosides (see Aminoglycoside Dosing Guideline)

II. Micafungin dose adjustment could be considered for patients with BMI > 35 when treating disseminated or severe life-threatening candidiasis. Dose should be increased from 100 mg to 150 mg daily.

BMI is automatically calculated in MiChart (Flowsheets tab-> Height/Weight).
BMI can also be calculated using the following equation:

$$BMI = \frac{wt\ (in\ kg)}{ht\ (in\ m)^2}$$

$$Ideal\ body\ weight\ for\ men\ (kg) = 2.3 * (ht\ in\ inches - 60) + 50$$

$$Ideal\ body\ weight\ for\ women\ (kg) = 2.3 * (ht\ in\ inches - 60) + 45.5$$

$$Adjusted\ body\ weight\ (kg) = (actual\ body\ weight - \ ideal\ body\ weight) * 0.4 + ideal\ body\ weight$$

| **Last revised:** 01/2025 | **Antimicrobial Stewardship Subcommittee:** 01/2025 | **P&T:** 01/2025 |
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**References**


| | Citation |
|---|---|
| 1 | Polso AK et al. Impact of hospital guideline for weight-based antimicrobial dosing in morbidly obese adults and comprehensive literature review. J Clin Pharm Ther 2014;39:584- 608. |
| 2 | Turner RB et al. Prospective, Controlled Study of Acyclovir Pharmacokinetics in Obese Patients. Antimicrob Agents Chemother 2016;60:1830- 1833. |


**Revision History**

| Date | Revision |
|---|---|
| 10/2019 | Added ribavirin and cidofovir |
| 08/2021 | Added flucytosine |
| 02/2025 | Added Micafungin |


*The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time-to-time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source*